Scheme of Valuation

QP Code: 25900356

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECT203 - INTEGRATED ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 mark

1. c). EB reverse-biased, CB reverse-biased
2. b). 00-01-10-11
3. c). V/μs
4. a). The voltage required between inverting and non-inverting inputs to make output zero
5. b). IC 741
6. b). 3 layers, 2 junctions
7. a). Twisted ring
8. d). All of the above
9. b). Less than the open-loop gain
10. b). Propagation delay

Part B

Short Answer Questions

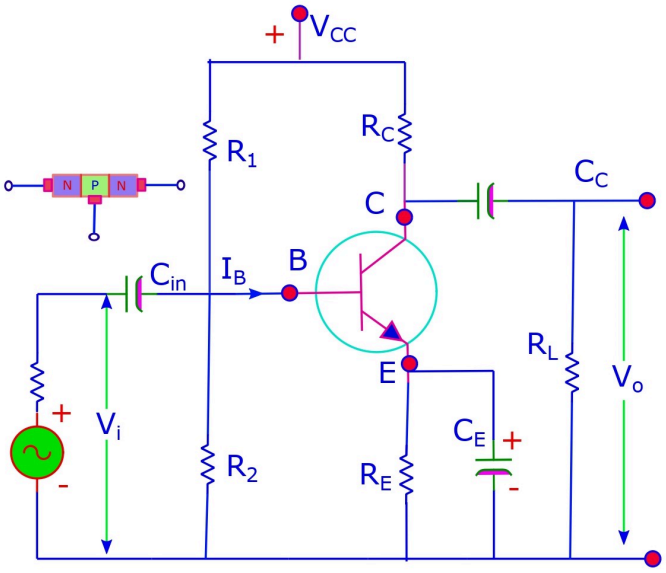
Answer any 4 questions

Each question carries 5 marks

11. Define a Single-stage CE Amplifier and sketch its circuit with voltage divider biasing.

Definition - 2 Marks

Circuit Diagram with proper labeling - 3 Marks

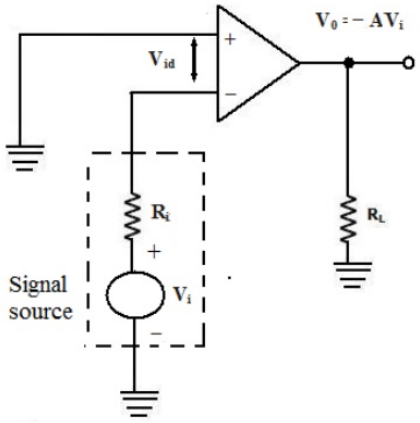


12. Draw the block diagram of a Parallel-In Serial-Out (PISO) shift register and explain the role of the control input (shift/load).

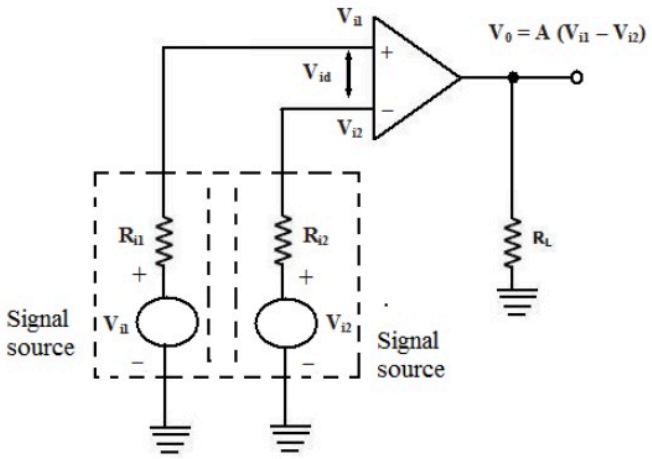
Block Diagram - 2 Marks
Explanation of the role of the control input Shift/ Load - 3 Marks

13. List and briefly explain any two op-amp open-loop configurations.

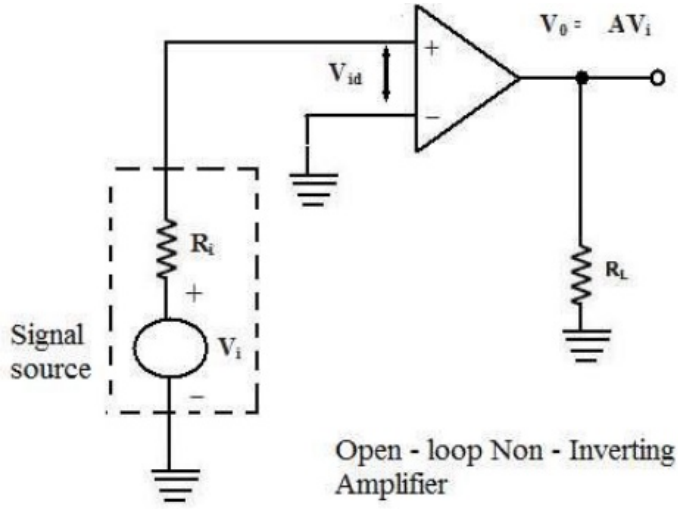
Circuit Diagram of any two open-loop configurations - 3 Marks (1.5 each)
Explanation - 2 Marks (1 each)



Open - loop Inverting Amplifier



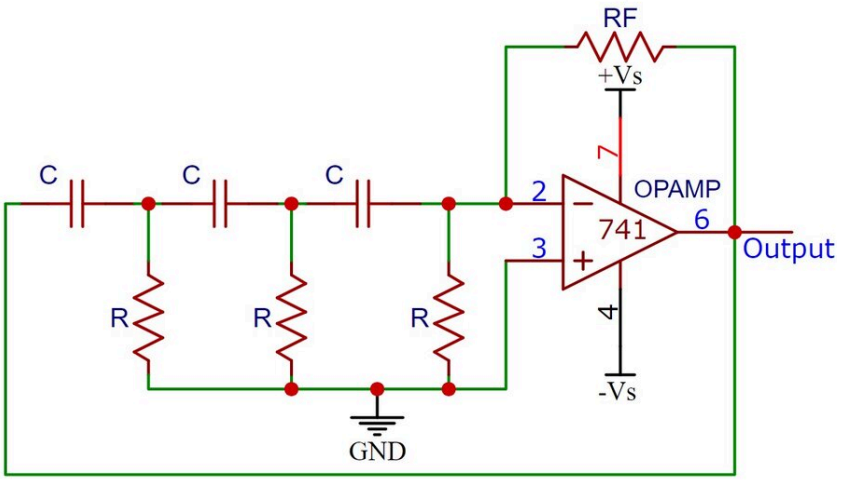
Open - loop Differential Amplifier



Open - loop Non - Inverting Amplifier

14. Explain the principle of RC phase shift oscillator using Op-Amp.

Circuit diagram - 2 Marks
Working Principle - 2 Mark
Frequency of oscillation - 1 Mark



Frequency = $= \frac{1}{2\pi RC\sqrt{6}}$

15. Explain the role of timing resistor and capacitor in IC 566.

Function of 566 - 1 Mark
Role of timing resistor - 2 Marks
Role of timing Capacitor - 2 Marks

The IC 566 is a voltage-controlled oscillator (VCO) that generates a square wave and a triangular wave. Its frequency of oscillation depends on an external timing resistor and timing capacitor, along with the applied control voltage.

Role of Timing Resistor :

- The timing resistor determines the charging and discharging current of the external capacitor.
- This current sets how fast the capacitor voltage rises and falls between the two threshold voltages inside the IC.

- A higher R means lower charging current → lower frequency,
  while a lower R gives higher frequency.

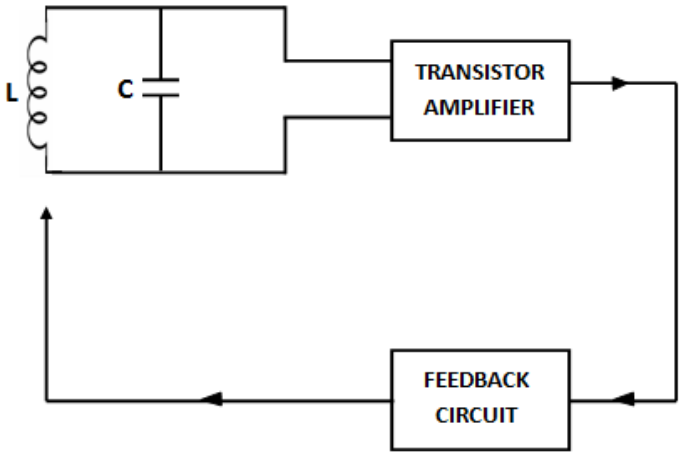
Role of Timing Capacitor (C) :

- The timing capacitor stores and releases charge during the oscillation cycle.
- Its charging and discharging time (through R) determines the oscillation period.
- Increasing C increases the charge/discharge time, thus decreasing the frequency of oscillation.

16. Discuss the essentials of an oscillator.

Definition of an Oscillator - 1 Mark
Explanation of Essentials of an oscillator with diagram- 4 Marks

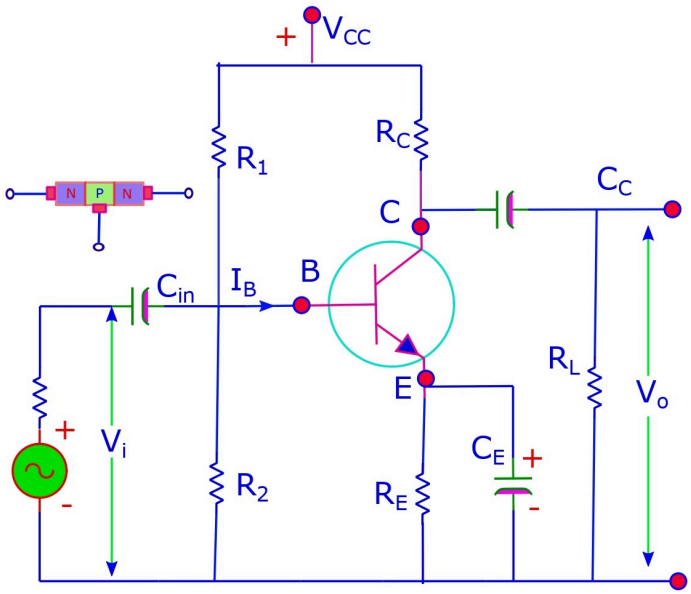
An oscillator is an electronic circuit that generates a continuous, periodic, and oscillating output signal without any external input signal.
It converts DC power into AC output at a desired frequency.



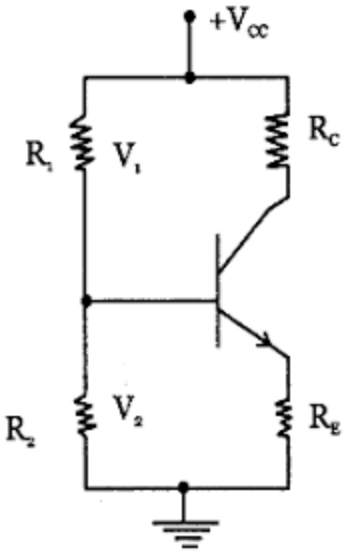
Part C
Essay Questions
Answer any 2 questions
Each question carries 10 marks

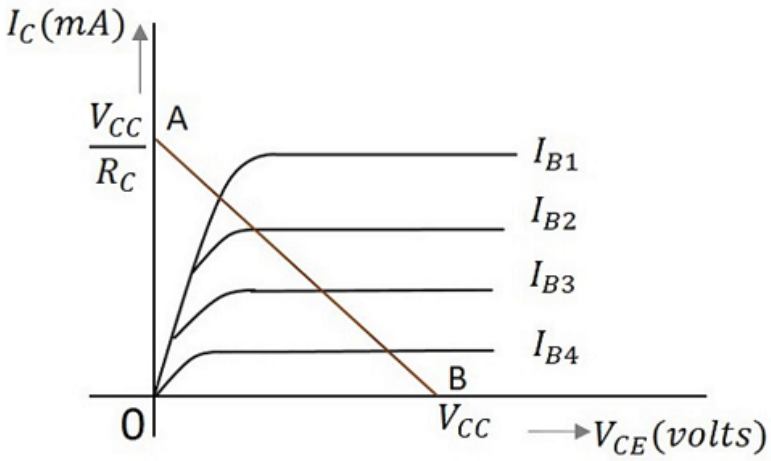
17. Draw and explain the dc equivalent circuit and dc load of a single-stage CE amplifier.

Circuit diagram of a Single Stage CE amplifier - 2 Marks
DC Equivalent circuit - 2 Marks
Explanation of DC equivalent - 2 Marks
DC Load line analysis with explanation - 4 Marks



A single-stage Common Emitter (CE) amplifier is the most widely used transistor amplifier configuration.It provides high voltage gain and moderate current gain.
To analyze and bias the transistor properly, we use the DC equivalent circuit and the DC load line to establish the quiescent operating point (Q-point).

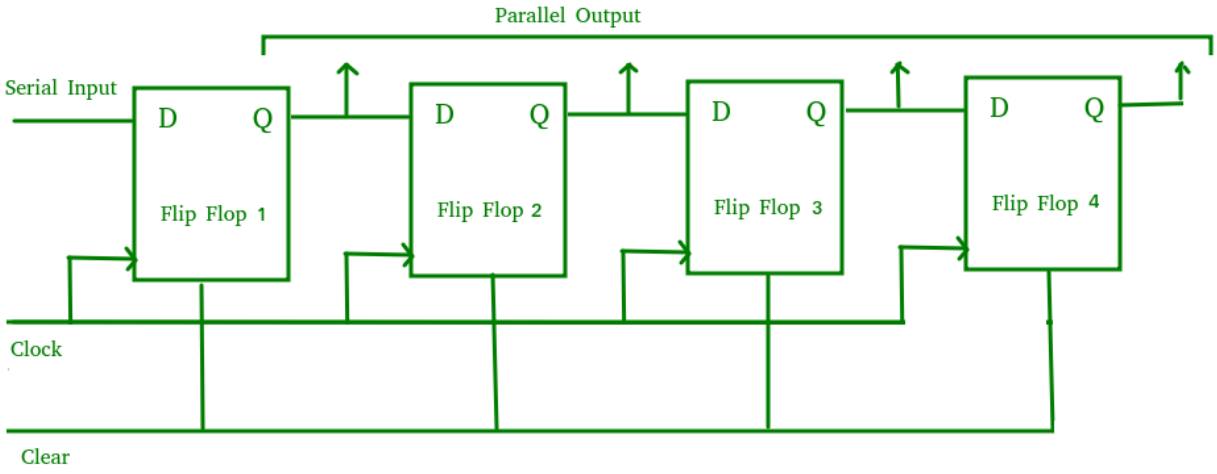




18. Design a 4-bit Serial-In Parallel-Out (SIPO) register using D flip-flops. Show the output sequence for the input 1011.

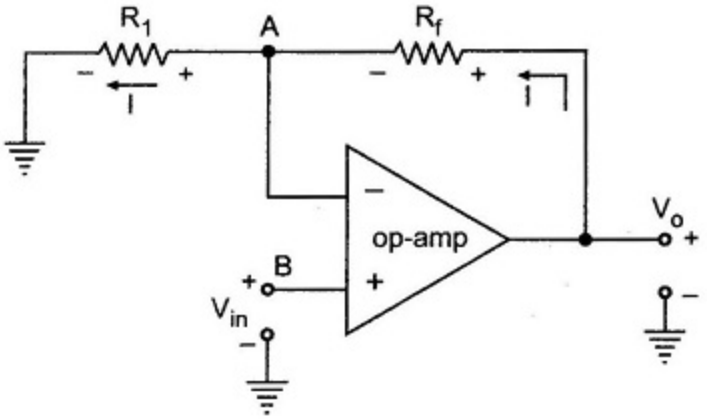
Definition - 1 Mark
Circuit Diagram with working principle - 5 Marks
Illustration of the output sequence - 4 Marks

A Serial-In Parallel-Out (SIPO) shift register accepts data serially (one bit per clock pulse) and makes all the bits available simultaneously (in parallel) at its outputs after the required number of clock pulses.



19. Derive the expression for the voltage gain of a closed-loop non-inverting amplifier.

Circuit Diagram - 4 Marks
Derivation with necessary steps - 6 Marks



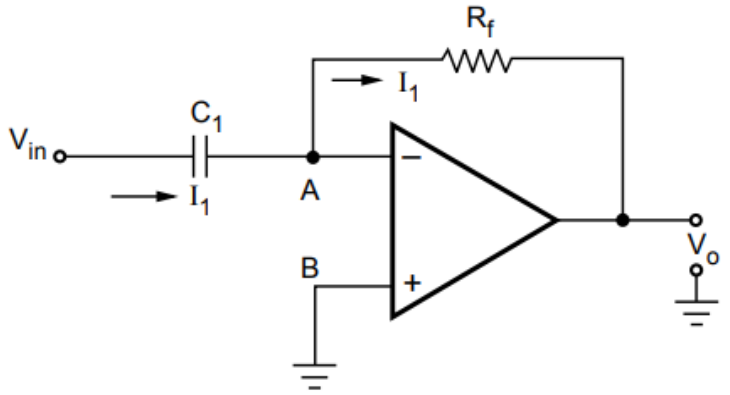
$$A_{VF} = \frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_1}$$

20. With the circuit diagram and derivation, explain the working of a *differentiator.*

Circuit Diagram - 2 Marks

Working Principle - 2 Marks
Derivation - 4 Marks
Practical Differentiator - 1 Mark
Input -output waveform - 1 Mark

The circuit which produces the differentiation of the input voltage at its output is called a differentiator.



The node B is grounded. The node A is also at the ground potential hence $V_A = 0$.

As input current of op-amp is zero, entire current I flows through the resistance Rf.

From the input side we can write,

$$I_1 = C_1 \frac{d(V_{in} - V_A)}{dt} = C_1 \frac{dV_{in}}{dt}$$

From the output side we can write,

$$I = \frac{(V_A - V_o)}{R_f} = -\frac{V_o}{R_f}$$

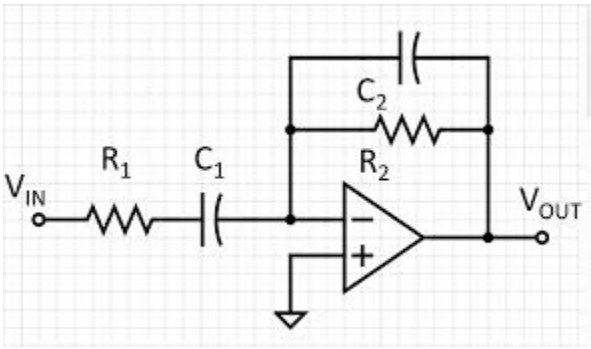
Equating the two equations,

$$C_1 \frac{dV_{in}}{dt} = -\frac{V_o}{R_f}$$

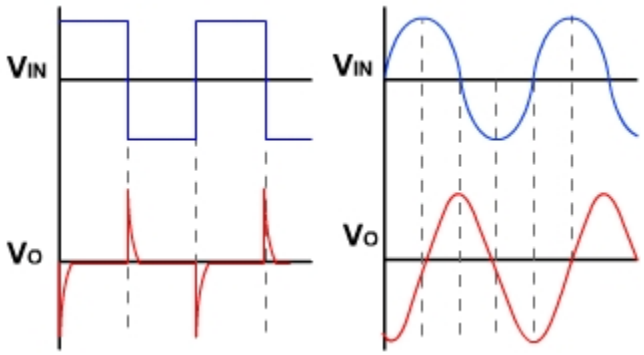
$$\boxed{V_o = -C_1 R_f \frac{dV_{in}}{dt}}$$

shows that the output is C_1R_f times the differentiation of the input, and the product C_1R_f is called the time constant of the differentiator.

The negative sign indicates that there is a phase shift of 180° between input and output.



Practical Differentiator



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